SARLANG-1M: A Benchmark for Vision–Language Modeling in SAR Image Understanding

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***Abstract*—Synthetic aperture radar (SAR) is a crucial remote sensing technology, enabling all-weather, day-and-night observation with strong surface penetration for precise and continuous environmental monitoring and analysis. However, SAR image interpretation remains challenging due to its complex physical imaging mechanisms and significant visual disparities from human perception. Recently, vision–language models (VLMs) have demonstrated remarkable success in RGB image understanding, offering powerful open-vocabulary interpretation and flexible language interaction. However, their application to SAR images is severely constrained by the absence of SAR-specific knowledge in their training distributions, leading to suboptimal performance. To address this limitation, we introduce *SARLANG-1M*, a large-scale benchmark tailored for multimodal SAR image understanding, with a primary focus on integrating SAR with textual modality. *SARLANG-1M* comprises more than 1 million high-quality SAR image–text pairs collected from over 59 cities worldwide. It features hierarchical resolutions (ranging from 0.1 to 25 m), fine-grained semantic descriptions (including both concise and detailed captions), diverse remote sensing categories (1696 object types and 16 land cover classes), and multitask question–answering pairs spanning seven applications and 1012 question types. Extensive experiments on mainstream VLMs demonstrate that fine-tuning with *SARLANG-1M* significantly enhances their performance in SAR image interpretation, reaching performance comparable to human experts. The dataset and code will be made publicly available at https://github.com/Jimmyxichen/SARLANG-1M**

***Index Terms*—High-quality texts, image captioning, synthetic aperture radar (SAR), vision–language models (VLMs), visual question–answering (VQA).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a vital remote sensing technology renowned for its ability to capture high-resolution images under all weather conditions, including clouds, rain, and darkness. Unlike optical imagery, SAR utilizes microwave signals that can penetrate materials such as vegetation, dry soil, and man-made structures, depending on the wavelength. Longer wavelengths, like L-band or P-band, enable SAR to penetrate forest canopies and reveal subsurface features, making it invaluable for forestry monitoring, archeological exploration, and soil moisture analysis [1], [2]. Combined with its all-weather and day–night imaging capabilities, SAR's penetration ability makes it an essential tool for applications such as disaster management, environmental monitoring, and military surveillance [3], [4]. These unique advantages have spurred growing research interest in advancing robust SAR image interpretation.

Despite significant advancements in various SAR-related tasks such as object detection [5], [6], semantic segmentation [7], [8], and other areas [9], SAR image interpretation remains highly challenging. One of the primary difficulties arises from the fundamental differences between SAR and optical imagery [10], [11]. Unlike RGB images, which capture visible light, SAR images are generated through microwave backscattering, where the radar system emits microwave pulses toward the Earth's surface and captures the reflected signals. This process introduces distinctive characteristics such as speckle noise and geometric distortions that complicate image analysis. Additionally, the inherent semantic complexity of SAR images—often characterized by intricate textures and nonintuitive patterns—makes interpretation more challenging than that of natural optical imagery. Moreover, the disparity between human visual perception and SAR representations further complicates annotation, requiring domain expertise and extensive training. Due to the high costs [12] and specialized knowledge required for annotation, existing SAR datasets remain limited in both size and diversity. These challenges underscore the need for specialized datasets, algorithms, and models tailored to SAR data, driving ongoing research and innovation in the field.

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Fig. 1. Exemplar cases of mainstream VLMs that exhibit strong performance on RGB images but struggle with paired SAR images. The first row demonstrates the image captioning results generated by the DeepSeekVL-7B [15], GPT-4o [16], and LLaVA1.5-7B [17] models, with the left, center, and right examples illustrating their respective outputs. The second row presents the VQA results produced by the same models, with the left, center, and right examples further highlighting the challenges associated with SAR image interpretation. Key terms in the text are highlighted with red underlines for emphasis. For instance, the LLaVA1.5-7B [17] model, as shown in the right example in the first row, misclassifies a residential scene as a person's hand. Furthermore, in the right example in the second row, the LLaVA1.5-7B [17] model fails to identify the presence of water in the SAR image, despite successfully recognizing it in the corresponding RGB image.

Recently, vision–language models (VLMs) [13], [14] have demonstrated significant potential in enhancing optical RGB image understanding through large-scale pretraining and instruction tuning. Unlike traditional models, which are often restricted to fixed vocabularies and predefined labels, VLMs exhibit superior image interpretation capabilities due to several key advantages. First, they enable flexible language interaction, allowing for more intuitive and context-aware image comprehension. Second, through multimodal pretraining on large-scale datasets, VLMs acquire extensive open-vocabulary language capabilities, enabling them to describe a wide range of scenes and concepts beyond predefined categories. The rapid advancement of VLMs has created new opportunities for SAR image interpretation. However, directly applying existing VLMs to SAR image analysis is impractical. These models are predominantly trained on RGB image datasets, meaning that SAR images fall outside the distribution of their training data. As a result, VLMs perform poorly when applied to SAR images, which exhibit fundamentally different data characteristics from RGB imagery. An example of this performance gap is illustrated in Fig. 1.

To enable VLMs to effectively interpret SAR images, the development of a large-scale SAR–text dataset is both essential and urgent. However, constructing such a dataset presents significant challenges due to data scarcity and the complexity of annotation. One of the primary obstacles is the lack of large-scale SAR image datasets with high-quality text annotations. Annotating SAR images requires expert-level supervision, as the unique characteristics of SAR images demand specialized domain knowledge for accurate interpretation. The high cost and complexity associated with SAR annotation have severely constrained the scale and diversity of existing datasets, thereby limiting the advancement of SAR-specific VLMs.

To overcome these challenges and advance SAR image understanding, this article introduces *SARLANG-1M*, a novel large-scale benchmark designed for multimodal SAR image interpretation, as illustrated in Fig. 2. The *SARLANG-1M* dataset consists of approximately 1 million high-quality SAR image–text pairs collected from over 59 cities worldwide. It encompasses multiscale resolutions ranging from 0.1 to 25 m, fine-grained semantic descriptions (including both concise and detailed captions), a diverse set of remote sensing categories (1696 object categories and 16 land cover categories), and multitask question–answering pairs covering seven applications and 1012 question types.

To generate high-quality and domain-specific text descriptions, we constructed the *SARLANG-1M* dataset using two distinct text generation approaches. The first approach involves text transfer through modality pairs. Specifically, we collected a large set of paired optical RGB and SAR images, where state-of-the-art VLMs (e.g., GPT-4o [16]) were used to generate high-quality text annotations for the optical RGB images. These annotations were then naturally aligned with their corresponding SAR images. To ensure accuracy, we manually reviewed and filtered out incorrect or irrelevant descriptions, such as references to color information that are applicable to RGB images but not to SAR images. This approach enables the transfer of knowledge from the RGB domain to SAR modalities, producing comprehensive and precise text annotations. However, existing VLMs exhibit limited spatial reasoning capabilities, particularly in identifying positional and quantitative attributes of objects—key aspects for remote sensing interpretation. To address this limitation, our second strategy involves generating fine-grained text annotations from existing bounding box annotations in SAR object detection datasets [18]. By combining this approach with rigorous manual verification by SAR experts, the *SARLANG-1M* dataset comprises 118 331 SAR images with 1 126 277 human-verified text annotations. This dataset facilitates the training and evaluation of VLMs across various SAR image understanding tasks, including visual question–answering (VQA) and image captioning.

Furthermore, we conduct comprehensive experiments using the *SARLANG-1M* dataset, evaluating both traditional models and ten state-of-the-art VLMs (e.g., DeepSeek-VL [15] and QWEN2.5-VL [19]), demonstrating the dataset's effectiveness in enhancing VLMs' ability to interpret SAR images. Specifically, *SARLANG-1M* achieves significant performance improvements, with increases of 67.20% in CIDEr [20] and 40.22% in GPT-4-based accuracy [21] for SAR image captioning and SAR VQA tasks, respectively. The visualization results of VLM's predictions across seven SAR image applications further validate the effectiveness of our *SARLANG-1M* dataset. Additionally, per-word Grad-CAM [22] visualizations are provided to demonstrate that *SARLANG-1M* enables VLMs to better align textual keywords with the visual content in SAR images.

The key contributions of this work are summarized as follows.

1) *Comprehensive Large-Scale Dataset: SARLANG-1M* consists of 118 331 multiresolution SAR images,

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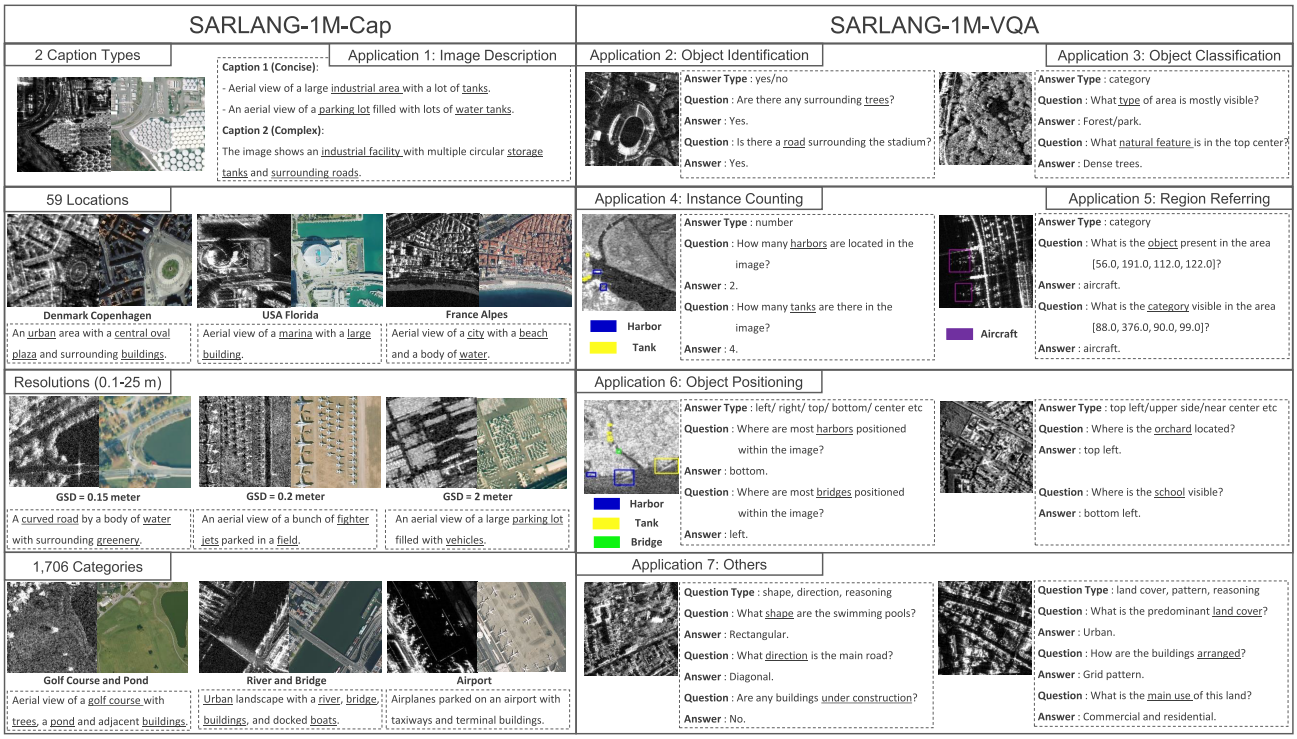


Fig. 2. Examples from the *SARLANG-1M* dataset. *SARLANG-1M* consists of two benchmarks: *SARLANG-1M-Cap* for SAR image captioning and *SARLANG-1M-VQA* for SAR image VQA. The *SARLANG-1M-Cap* benchmark is characterized by four dimensions: caption type, the presence of location information, resolution, and the number of categories, supporting the task of detailed SAR image description. In contrast, the *SARLANG-1M-VQA* benchmark is designed to enable six additional applications, including object identification, object classification, instance counting, region referring, object positioning, and others.

providing 1 080 627 high-quality image–text pairs and 45 650 detailed captions from over 59 cities worldwide. The dataset spans multiscale resolutions from 0.1 to 25 m and includes fine-grained semantic descriptions (both concise and detailed captions), diverse remote sensing categories (1696 object categories and 16 land cover types), and multitask question–answering pairs covering 1012 question types.

2) *Benchmark for Vision–Language Tasks:* The *SARLANG-1M* benchmark comprises two key components: *SARLANG-1M-Cap* for image captioning task and *SARLANG-1M-VQA* for VQA task. As the largest SAR image–text dataset to date, it supports seven critical remote sensing applications, including image description, object identification, object classification, instance counting, region referring, and object positioning.
3) *Extensive Model Evaluation and Improvement:* We conduct a comprehensive performance evaluation using *SARLANG-1M*, assessing two state-of-the-art traditional models and ten VLMs. The experimental results demonstrate that fine-tuning mainstream VLMs on *SARLANG-1M* significantly enhances their performance on SAR-specific vision–language tasks, achieving results comparable to those of human experts.

The remainder of this article is organized as follows. Section II reviews related work. Section III details the construction progress of the *SARLANG-1M* dataset. Section IV presents three VLM evaluation pipelines for SAR image understanding. The experimental results and analysis are provided in Section V. Finally, Section VII draws the conclusion.

II. Related Works

A. SAR for Perception

Early SAR image processing models typically relied on manually crafted features and simple statistical learning models to categorize these features into predefined classes or patterns [23], [24]. However, these approaches were heavily dependent on domain expertise for feature creation, rendering them less effective for complex tasks. As a result, the scalability and generalizability of these models were significantly constrained. The advent of deep learning, characterized by end-to-end trainable neural networks, has transformed SAR image analysis by eliminating the need for intricate feature extraction processes. This advancement has led to higher accuracy and enhanced model robustness. Prominent deep learning architectures such as convolutional neural networks (CNNs) [25], long short-term memory (LSTM) networks [26], Transformers [27], diffusion models [28], and the you only look once (YOLO) [29] framework have significantly advanced the field of SAR image understanding. These models have improved performance across various tasks, including classification [30], [31], [32], [33], semantic segmentation [34], [35], [36], object detection [37], [38], [39], [40], and image-to-image translation [41], [42], [43], [44].

B. Text Generation for Remote Sensing

1) Image Captioning: Remote sensing image captioning (RSIC) aims to describe key objects within remote sensing images using natural language. RSIC faces several challenges, including scale variations and cross-modal disparities. Objects

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belonging to the same category may appear in different scales within remote sensing images, and significant modality differences exist between the input image and the output text. This necessitates precise alignment and effective fusion of image and text features during text generation.

Current RSIC models can be categorized into three primary approaches: retrieval-based methods, syntax-template-based methods, and encoder–decoder-based methods. Retrieval-based methods attempt to generate captions by retrieving similar images and using their annotated sentences [45], [46]. Syntax-template-based methods involve detecting objects in images, identifying candidate words, and filling them into predefined syntax templates [47]. However, the sentences generated by these two methods often lack flexibility and naturalness.

Recently, encoder–decoder-based approaches have been developed to learn representations of remote sensing images and grammar automatically, allowing for the creation of more adaptable sentences. These models generally operate in two stages: vision feature extraction and text generation. In the vision feature extraction stage, CNNs [25] or Vision Transformers [48] are typically employed to extract semantic features from images and encode them into high-dimensional vectors. For text generation, recurrent neural networks (RNNs) [49] or Transformers [27] are used to transform these feature vectors into text. Efforts to enhance feature extraction include the work by Zheng et al. [50], who introduced attention-based feature fusion modules alongside an auxiliary object detection task to enrich visual feature representation. Similarly, Ye et al. [51] proposed a multilabel classification task to capture prior knowledge, complemented by a semantic gate module to guide the generation of hidden states.

In terms of enhancing the decoder, Hoxha and Melgani [52] developed a support vector machine (SVM)-based decoder capable of transforming visual vectors into high-quality textual descriptions, especially for small datasets. Furthermore, Transformer-based techniques have been explored: Chen et al. [53] constructed a Swin Transformer network to explore image features, using the Transformer network as a decoder to generate captions. Liu et al. [54] proposed a multilayer aggregated Transformer as the decoder, with LSTM networks linking the Transformer encoder and decoder for improved feature representation. Yang et al. [55] utilized a dual-LSTM architecture as the decoder for generating coherent and contextually relevant captions. Instead of merely concatenating generated visual features, word features, and LSTM hidden states for input into the LSTM network, a cross-modal feature alignment module is used to align visual features with other input features before fusion. Additionally, a cross-modal feature alignment loss is introduced to minimize the distance between encoded visual features and LSTM-encoded sentence features.

2) VQA: VQA for remote sensing (RSVQA) focuses on generating accurate answers to natural language inquiries by comprehensively analyzing the visual content of remote sensing images. The standard framework for addressing the RSVQA problem entails the integration of an aerial image with a corresponding natural language question to output a contextually relevant answer derived from the image's content. The types of questions typically encountered in RSVQA include, but are not limited to, inquiries regarding categories, quantities, characteristics, spatial positions, and purposes of key objects within the aerial scene, as well as the relative relationships between distinct objects. The research of RSVQA was initiated by Lobry et al. [56], who introduced a novel RGB–text dataset derived from Sentinel-2 images, alongside a deep learning framework to assess the dataset's efficacy. This framework is systematically organized into three distinct stages: feature extraction, feature fusion, and prediction based on the fused features. During the feature extraction phase, a CNN-based model is utilized to extract visual features from the input remote sensing images, while an RNN-based model is employed to embed a sentence of words into a latent space, thereby generating a linguistic embedding. Specifically, ResNet-152 [57], pretrained on ImageNet [58], serves as the visual encoder, while the skip-thought model [59], trained on the BookCorpus dataset [60], functions as the language encoder. The extracted visual feature and textural embedding are subsequently fused via pointwise multiplication in the feature fusion stage, followed by a projection onto a fixed-dimensional vector using a multilayer perceptron (MLP) layer. The follow-up works focus on enhancing this framework by substituting the convolutional and recurrent encoders with more advanced architectures. For instance, Silva et al. [61] integrated the multimodal medical bidirectional encoder representations from transformers (MMBERT) [62] with an EfficientNetV2 [63] image encoder and a RealFormer [64] multimodal encoder. Similarly, Tosato et al. [65] embed an attention mechanism [27] guided by segmentation into the RSVQA pipeline, positing that segmentation critically guides attention by delivering a contextual comprehension of the visual information, thereby emphasizing specific objects or areas of interest. In their work, the ResNet-50 [57] and DistilBERT [66] models are leveraged as the visual and textual encoders, respectively.

C. Vision–Language Models

1) General VLM: Foundation models have recently emerged as a central topic in AI research due to their ability to address a variety of downstream tasks after undergoing extensive pretraining. A vision foundation model (VFM) specifically targets visual tasks, while a VLM enhances VFMs by integrating both visual and textual modalities, facilitating more complex multimodal reasoning and a wider range of applications. The rise of expansive large-scale image–text datasets has propelled VLMs into the spotlight, as they have demonstrated exceptional performance on multimodal tasks such as image captioning, VQA, and cross-modal retrieval. Typically, mainstream VLMs comprise a pretrained visual encoder to process visual data, a language encoder to interpret user instructions and generate responses, and a vision–language cross-modal connector to fuse visual feature representations with textual embeddings. A key component in pretraining VLMs is the effective bridging of vision and language through image–text pairs. This is generally accomplished by employing two primary objectives: contrastive learning and generative modeling.

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TABLE I

DETAILED INFORMATION OF FOUR PUBLIC DATASETS USED TO CONSTRUCT OUR SARLANG-1M DATASET, INCLUDING POLARIZATIONS, RESOLUTION, IMAGE SIZE, COLLECTED SENSORS, AND SAR IMAGE NUMBER. "# *P*" DENOTES THE POLARIZATIONS. "IMAGE SIZE" IS REPRESENTED AS THE FORM OF "WIDTH/HEIGHT"

Dataset	# P	Resolution (m)	Image Size	Sensor	# SAR Image
SpaceNet6 [83]	4	0.5	900 / 455 $\sim$ 795	Capella Space's Satellite	3,311
DFC2023 [84]	1	0.5, 0.8, 1, 2	512 / 512	SuperView-1 ("GaoJing" in Chinese) Gaofen-2, Gaofen-3 Satellites	5,222
OpenEarthMap-SAR [85]	1	0.15 $\sim$ 0.5	1024 / 1024	Umbra Satellites	4,813
SARDet-100k [18]	1	0.1 $\sim$ 25	512 / 512	Gaofen-3, Sentinel-1, TerraSAR-X RadarSat-2, TanDEMX, HISEA-1 Airborne SAR Synthetic Slice	104,985

Contrastive learning, exemplified by the CLIP [67] model, is designed to align image–text pairs by maximizing the similarity between matched pairs while minimizing it for mismatched ones. This methodology allows CLIP to develop a unified representation space for both visual and textual data, leading to outstanding zero-shot classification capabilities. Following this, OpenCLIP [68] offers a fully open-source version of the CLIP model. Building on CLIP's architecture, EVA-CLIP [69] introduces enhanced training strategies aimed at improving performance, reducing computational overhead, and increasing training stability.

Generative modeling focuses on training models to produce coherent and relevant text or images, primarily through two methods: masked reconstruction and autoregressive next-token prediction. Masked reconstruction, employed by models like FLAVA [70] and MaskVLM [71], involves predicting masked tokens within text or occluded patches in images. This technique enhances the model's ability to understand context and relationships across visual and linguistic modalities, thereby improving cross-modal comprehension. Autoregressive next-token prediction, a dominant paradigm in VLM training, aims to generate the next token in a sequence using prior context. This approach typically involves three core components: a pretrained language module such as Llama3 [72] or Vicuna [73], a pretrained visual encoder like EVA-CLIP [69] or CLIP [67], and a trainable connection module to bridge visual and language embeddings, such as MLP layers [74] or Q-Former [75]. This architecture allows the model to leverage pretrained visual and language capabilities while the connection module learns to align information between these modalities. LLaVA [17] exemplifies pioneering work in this area by integrating and surpassing the strengths of both Vicuna [73] and CLIP. It achieves this by connecting CLIP's open-set visual encoder with Vicuna's language decoder and then fine-tuning end-to-end on generated instructional vision–language data. Additionally, other approaches [75], [76], [77] are designed to enhance vision–language representation learning by employing multiple training objectives. For instance, BLIP [76] employs a pretraining strategy akin to CLIP but utilizes distinct training objectives. Specifically, BLIP incorporates contrastive learning to align visual and textual information by pulling together corresponding image–text pairs and pushing apart noncorresponding ones. However, BLIP also employs techniques such as image–text matching and captioning objectives to enhance its learning process, thus improving the representations derived from image–text pairs. This multiobjective strategy allows BLIP to capture rich and nuanced relationships between images and text beyond the scope of contrastive learning alone.

2) VLM for Remote Sensing: Recently, VLMs [14], [78] have been applied to the field of remote sensing imagery, demonstrating significant potential in addressing various downstream tasks such as zero-shot classification, image captioning, VQA, and object referencing. Building on the CLIP [67] architecture, Liu et al. [78] collected a comprehensive remote sensing dataset comprising 17 subdatasets to pretrain the RemoteCLIP [78] model, subsequently evaluating its performance across diverse downstream tasks. In contrast to the CLIP [67] model, RemoteCLIP [78] employs the InfoNCE loss function to calculate the similarity between visual and textual features. Similarly, Kuckreja et al. [14] developed a remote sensing multimodal instruction-following dataset that includes images with brief descriptions generated by Vicuna-v1.5 [73]. Using this dataset, Kuckreja et al. [14] fine-tuned the LLaVA-1.5 [17] model with low-rank adaptation (LoRA) [79] technology. Recognizing the capability of MiniGPT-4 [80] in training a single projection layer to effectively align visual features with large language models (LLMs), RSGPT [13] was developed by fine-tuning InstructBLIP [77] on their RSICap dataset. By fine-tuning only the Q-Former [75] network and the linear layer of InstructBLIP, the RSGPT model efficiently learns to align visual features of remote sensing images with LLMs in a data-efficient manner. The advancement of VLMs for remote sensing imagery has been facilitated by the availability of large-scale RGB image–text datasets [21], [78]. However, to the best of authors' knowledge, there is a scarcity of SAR image–text datasets due to the high cost and complexity of SAR annotation, which significantly limits the development of VLMs in the SAR image domain. A more systematic review could be found in recent surveys [81], [82].

III. SARLANG-1M DATASET

A. Dataset Description

In this section, we first provide an overview of the *SARLANG-1M* dataset, detailing its geographic coverage, sensor types, resolution scales, and category distribution. Additionally, we further present a comparative analysis between *SARLANG-1M* and existing SAR interpretation datasets, highlighting its unique advantages and contributions to the field.

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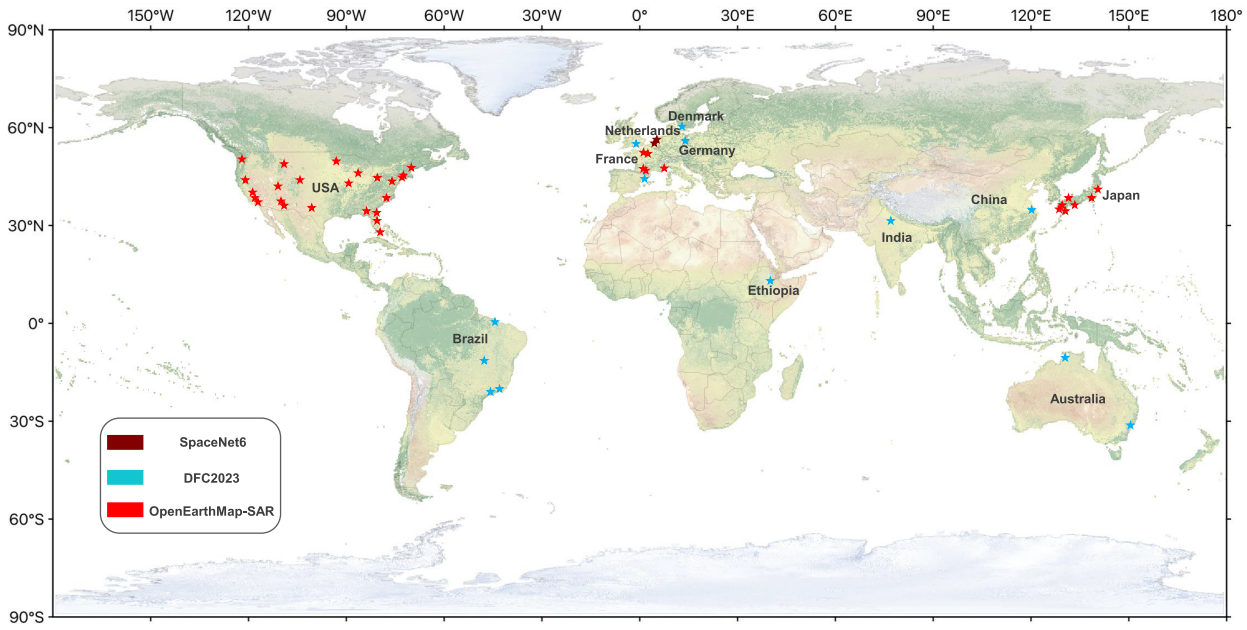


Fig. 3. Cities included in the *SARLANG-1M* dataset. *SARLANG-1M* contains SAR images from over 59 cities worldwide, most of which are highlighted on the map.

TABLE II

COMPARISON WITH EXISTING SAR DATASETS IN TERMS OF SAMPLE NUMBERS, RESOLUTIONS, SUPPORTED TASKS, AND TEXT AVAILABILITY

Dataset	Proposed Time	# Samples	Resolution (m)	Task	Provide Texts
OpenSARShip [86]	2018	11,346	10	Classification	✗
FUSAR-Map [87]	2021	610	3	Semantic Segmentation	✗
SSDD [88]	2021	1,160	1 ∼ 15	Ship Detection	✗
MSAR [89]	2022	28,449	≤ 1	Object Detection	✗
SADD [90]	2022	2,966	0.5 ∼ 3	Airplane Detection	✗
SAR-AIRcraft [91]	2023	4,368	1	Object Detection	✗
SIVED [92]	2023	1,044	0.1, 0.3	Object Detection	✗
SARDet-100k [18]	2024	104,985	0.1 ∼ 25	Object Detection	✗
BRIGHT [93]	2025	4,246	0.3 ∼ 1	Change Detection, Image Matching, SSL, UDA	✗
OpenEarthMap-SAR [85]	2025	5,033	0.15 ∼ 0.5	Semantic Segmentation, Image Translation, UDA	✗
Ours	2025	118,331	0.1 ∼ 25	Image Captioning, VQA	✓

1) Dataset Overview: *SARLANG-1M* consists of 118 331 SAR images, covering more than 59 cities worldwide, ensuring diverse geographic representation, as shown in Fig. 3. The dataset integrates SAR images from four distinct sources, collected by over 12 different satellites, as summarized in Table I. These images capture a broad range of common remote sensing environments, including airports, harbors, rivers, schools, residential areas, forests, and commercial zones, making it highly representative of real-world SAR imaging scenarios.

To support advancements in SAR image understanding under the VLM era, *SARLANG-1M* consists of two parts that support two key and different tasks: 1) *SARLANG-1M-Cap* for SAR image captioning and 2) *SARLANG-1M-VQA* for SAR image VQA. The SAR image captioning benchmark provides two levels of textual descriptions: *concise captions*, which offer brief but informative summaries of image content, and *detailed captions*, which provide richer semantic descriptions with more contextual information. The *SARLANG-1M-VQA* benchmark is structured into six primary question types: *object identification, object classification, instance counting, region referring, object positioning*, and *general inquiries*. These tasks are designed to challenge and evaluate the capabilities of VLMs in SAR-specific interpretation, ensuring a comprehensive assessment across various remote sensing applications.

2) Comparison With Existing Datasets: Table II presents a comprehensive comparison between *SARLANG-1M* and existing SAR datasets, highlighting its unique advantages. As shown in the table, *SARLANG-1M* offers a significantly

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TABLE III

COMPARISON OF EXISTING REMOTE SENSING DATASETS IN TERMS OF IMAGE AND TEXT SIMILARITIES. "# IMAGE" DENOTES THE NUMBER OF INCLUDED IMAGES. "IMAGE SIM." AND "TEXT SIM." REPRESENT THE SIMILARITY OF IMAGE AND TEXT, RESPECTIVELY, WHICH ARE REPRESENTED AS THE FORM OF "MEAN/VARIANCE." LOWER MEAN OR VARIANCE INDICATES HIGHER DATASET DIVERSITY. "–" MEANS NOT AVAILABLE

Modal	Dataset	# Image	Image Sim.	Text Sim.
RGB	RSICD [96]	10, 921	0.14 / 0.05	0.27 / 0.16
	RSITMD [97]	4, 743	0.14 / 0.05	0.20 / 0.13
	UCM [98]	2, 100	0.13 / 0.05	0.22 / 0.17
	VRSBench [21]	29, 614	0.13 / 0.04	0.48 / 0.12
	Ours (paired RGB)	13, 346	0.16 / 0.06	–
SAR	FUSAR-Map [87]	610	0.28 / 0.09	–
	SARDet-100k [18]	10, 4985	0.26 / 0.13	–
	Ours (SAR)	118, 331	0.25 / 0.06	0.49 / 0.16

broader range of spatial resolutions and a substantially larger number of annotated samples compared to previous datasets. One of the key distinguishing features of *SARLANG-1M* is the inclusion of high-quality text annotations, which enable advanced SAR image captioning and VQA tasks. This rich textual information bridges the gap between vision and language in SAR interpretation, providing a valuable resource for training and evaluating multimodal models.

To enable a fair and objective comparison, we introduce *overall similarity* as a metric to assess the redundancy and diversity of *SARLANG-1M* in relation to existing SAR datasets. The *overall image similarity* metric quantifies dataset quality by measuring the internal similarity of images, providing insights into the dataset's distributional uniqueness. Specifically, we compute the pairwise cosine similarity of image features for all image pairs within the dataset and aggregate these values into a similarity set. The *overall similarity* is characterized by two statistical measures: mean and variance. The mean similarity represents the central tendency of the dataset's feature similarity, with higher values indicating lower diversity due to greater redundancy among images. Meanwhile, the variance captures the dispersion of similarity values, reflecting the heterogeneity of the dataset. A lower variance suggests a more uniform distribution, whereas a higher variance indicates a greater mix of unique and similar samples. In our implementation, image features are extracted using the pretrained VGG16 model [94]. A similar approach is used to compute *overall text similarity*, where text features are extracted using the pretrained Sentence-BERT model [95]. This ensures a consistent methodology for evaluating both image and text redundancy across the dataset. The specifics of the *overall image similarity*, along with the *overall text similarity*, are provided in Appendixes A and B, respectively.

Table III presents a comparative analysis of our dataset with existing SAR datasets in terms of image similarity. In contrast to other SAR datasets [18], [87], our dataset demonstrates lower image similarity, indicating a higher degree of diversity and reduced redundancy among the SAR images in our collection. Furthermore, since existing SAR datasets lack text samples, we extend our comparison to include established RGB–text datasets, as detailed in Table III. To ensure a fair evaluation of image similarity, we compute the similarity metric using a subset of RGB images derived from the RGB–SAR pairs in our dataset. This approach guarantees that the comparison is conducted within the same modality. The results reveal that the image similarity of our dataset is comparable to the mainstream remote sensing datasets containing RGB–text pairs.

TABLE IV

SEVEN SAR APPLICATIONS SUPPORTED BY THE SAR–TEXT SAMPLES IN OUR SARLANG-1M DATASET. "CAP" AND "VQA" DENOTE THE SARLANG-1M-CAP BENCHMARK AND SARLANG-1M-VQA BENCHMARK, RESPECTIVELY

Application	*SARLANG-1M* Dataset	# SAR-text Sample
Image Description	Cap	45,650
Object Identification	VQA ('object identification' type)	484,620
Object Classification	VQA ('object classification' type)	132,525
Instance Counting	VQA ('instance counting' type)	117,382
Region Referring	VQA ('region referring' type)	221,450
Object Positioning	VQA ('object positioning' type)	106,171
Others	VQA ('general inquiries' type)	18,479

3) Supported Subtasks: As illustrated in Fig. 2, the text annotations in our *SARLANG-1M* dataset, including semantic captions and VQA labels, support seven applications: image description, object identification, object classification, instance counting, region referring, object positioning, and others. Table IV illustrates the relationship between dataset samples and each application, followed by the task definitions.

1) *Image Description:* This application involves structured parsing of satellite imagery by detailing key objects and their main characteristics. As shown in Table IV, the text annotations for image description primarily originate from the *SARLANG-1M-Cap* benchmark.
2) *Object Identification:* This application evaluates the model's SAR image recognition abilities by providing Boolean answers to the presence of specific categories. The annotations mainly correspond to "object identification" questions in the *SARLANG-1M-VQA* benchmark.
3) *Object Classification:* This application focuses on identifying visible object categories within SAR images. The corresponding text annotations in our *SARLANG-1M-VQA* benchmark encompass over 1696 remote sensing object categories.
4) *Instance Counting:* For a specified object category, this application requires the model to quantify instances within the SAR image.
5) *Region Referring:* Given a specific location in the SAR image, this application challenges the model to determine the category present in the local area. The "region referring" samples in our *SARLANG-1M-VQA* benchmark facilitate this application.
6) *Object Positioning:* This application assesses the model's capability to predict potential areas for specific object categories within SAR images. The "object positioning" samples in our *SARLANG-1M-VQA* benchmark offer precise regions for these categories.
7) *Others:* This category encompasses a variety of common tasks in the remote sensing domain, including

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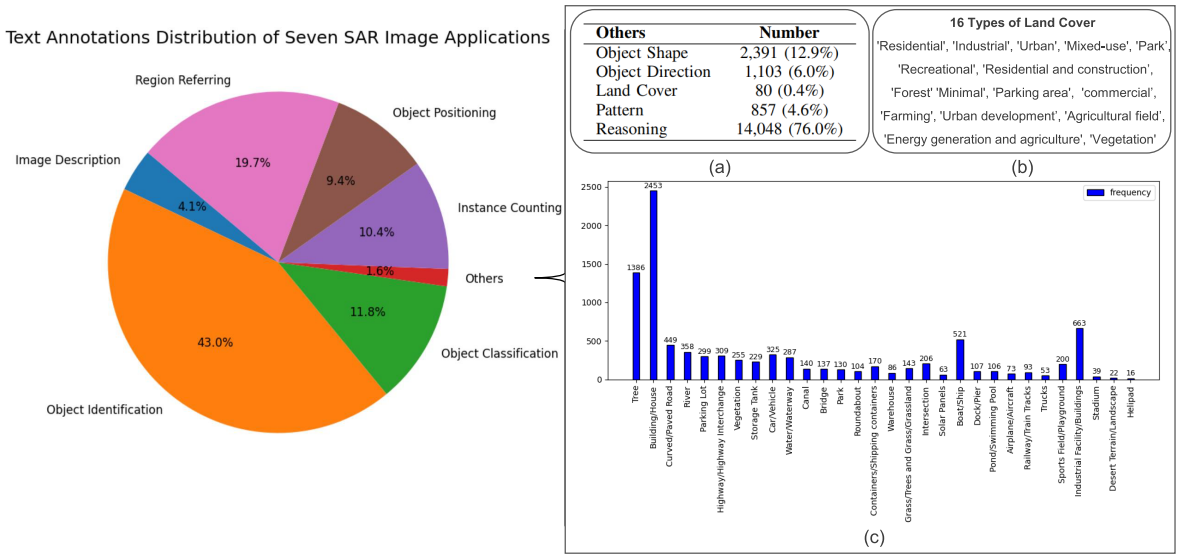


Fig. 4. Statistics of text annotations in *SARLANG-1M* benchmark. (a) Distribution of seven applications provided in *SARLANG-1M* benchmark. (b) Numbers of each question type in the "others" application. (c) Distribution of the 30 most frequent object categories.

object shape prediction, object direction prediction, land cover classification, pattern prediction, and reasoning functions. The "general inquiries" samples in our *SARLANG-1M-VQA* benchmark facilitate this application. Detailed examples for each function are shown in Fig. 2.

Our *SARLANG-1M* dataset comprises a total of 1 126 277 samples. The distribution of these samples across the seven applications is displayed on the left side of Fig. 4. Among these, the "others" category contains a variety of text annotations although they represent a smaller proportion. These annotations include five types of questions: object shape, direction, land cover, pattern, and reasoning. The statistics for each question type are listed in Fig. 4(a). Notably, "land cover" questions support a critical task in the remote sensing field, namely, land cover classification. The "land cover" item provides 80 text annotations and 16 land cover categories, as shown in Fig. 4(b). Additionally, our *SARLANG-1M* benchmark includes over 1696 object categories. Fig. 4(c) displays the distribution of the 30 most frequent remote sensing categories. Some exhibit nuanced vocabulary in textual expressions. For instance, the "Sports Field/Playground" category includes "Soccer Field," "Sports Field," "Tennis Courts," "Golf Course," and "Baseball Field." This diversity extends beyond the six categories in the object detection dataset [18], enabling broader support for remote sensing applications.

B. Dataset Construction

In this section, we first provide a concise overview of the data sources that form the foundation of the *SARLANG-1M* dataset. We, then, describe the generation process of high-quality textual annotations. Next, we outline the two different text generation pipelines employed for SAR image captioning and VQA tasks. Finally, we highlight key aspects of *SARLANG-1M*, including its seven downstream applications and the statistical characteristics of its textual annotations.

1) Data Source: The SAR images in *SARLANG-1M* are sourced from four publicly available SAR datasets: SpaceNet6 [83], DFC2023 [84], OpenEarthMap-SAR [85], and SARDet-100k [18]. Table I provides a summary of these datasets, detailing their key characteristics and acquisition parameters. Notably, the former three [83], [84], [85] provide strictly co-registered RGB–SAR image pairs, with the RGB images being cloud-free and haze-free to ensure the reliability of the transferred captions. The SARDet-100k [18] dataset includes SAR images and annotations of bounding boxes and corresponding class categories.

a) SpaceNet6 Dataset: The SpaceNet6 dataset [83] integrates half-meter quad-polarized X-band SAR images with corresponding half-meter optical imagery, focusing on the Port of Rotterdam, The Netherlands. The SAR images were acquired using an aerial sensor operated by Capella Space, which captured the region over three consecutive days: August 4, 23, and 24, 2019. The dataset consists of 204 individual SAR image strips, each containing quad-polarized data (HH, HV, VH, and VV) in the X-band wavelength. The imagery was collected from an off-nadir perspective, with relative look angles ranging from 53.4° to 56.6°, and includes observations from both north- and south-facing acquisition directions.

b) DFC2023 Dataset: The DFC2023 dataset [84] provides paired RGB and SAR images collected from multiple high-resolution satellites, including SuperView-1 (known as GaoJing in Chinese), Gaofen-2, and Gaofen-3, offering spatial resolutions of 0.5, 0.8, and 1 m, respectively. Additionally, normalized digital surface models (nDSMs) are provided as reference data, generated from stereo imagery captured by Gaofen-7 and WorldView satellites, with a ground sampling distance (GSD) of 2 m. The dataset spans 17 cities across six continents, encompassing multiresolution SAR images that enhance its geographic diversity.

c) OpenEarthMap-SAR Dataset: The OpenEarthMap-SAR dataset [85] comprises 1.5 million segmented samples derived from 5033 aerial and satellite images, each with a

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1024×1024 pixel resolution. These images cover 35 regions across Japan, France, and the United States. The dataset includes a mix of partially manually annotated and fully pseudo-labeled eight-class land cover classifications, with a GSD ranging from 0.15 to 0.5 m. The SAR samples in OpenEarthMap-SAR are sourced from Umbra, acquired in spotlight mode, with a resolution varying between 0.15 and 0.5 m. The optical data contain red, green, and blue spectral bands, while the SAR data primarily consist of amplitude information in VV or HH polarization bands. To ensure precise alignment between optical and SAR datasets, multiple experts manually co-registered the paired images and conducted cross-verification to meet strict quality standards.

d) SARDet-100k Dataset: The SARDet-100k dataset [18] is a large-scale SAR object detection dataset, comprising 104 985 images spanning six major object categories: ship, tank, aircraft, bridge, car, and harbor. The dataset aggregates SAR images from ten publicly available datasets, captured by seven different satellites, including Gaofen-3, Sentinel-1, TerraSAR-X, and RadarSat-2. Each SAR image in SARDet-100k is annotated with precise bounding boxes, formatted as (category, [$x_\min, y_\min$, width, height]), where $x_\min$ and $y_\min$ denote the coordinates of the top-left corner of the bounding box, while category specifies the object category contained within the bounding box. Width and height refer to the width and height of the bounding box in the SAR images, respectively. This dataset provides a rich resource for SAR-based object detection, with extensive annotations that enhance the training and evaluation of deep learning models.

These four publicly available datasets play distinct roles in constructing the *SARLANG-1M* dataset. SpaceNet6 [83], DFC2023 [84], and OpenEarthMap-SAR [85] contribute to both the *SARLANG-1M-Cap* and *SARLANG-1M-VQA* benchmarks, while SARDet-100k [18] is used exclusively for the *SARLANG-1M-VQA* benchmark. To ensure data integrity, SAR images with incomplete or missing information were removed from the original datasets. After this refinement, the *SARLANG-1M-Cap* benchmark comprises 13 346 SAR images, while the *SARLANG-1M-VQA* benchmark includes 118 331 SAR images.

2) Text Annotation Generation: The construction of large-scale text data generally follows two primary methods. The first method [99], [100] involves using pretrained foundation models to generate text based on specified prompts. While this approach could inadvertently introduce noisy labels, existing methods [80] often incorporate manual verification to significantly reduce the proportion of erroneous labels. The second method [13], [14] involves developing text templates to produce text, thereby avoiding the introduction of uncontrolled textual errors. However, the diversity of the generated text is constrained by the variety of the text templates.

Following the aforementioned principles, we employ two distinct text generation strategies to create high-quality textual descriptions for the *SARLANG-1M* benchmark, covering both *SARLANG-1M-Cap* and *SARLANG-1M-VQA*. For *SARLANG-1M-Cap*, we adopt a modality transfer approach, where textual descriptions are first generated for RGB images and then aligned with their corresponding SAR images. Since the paired RGB and SAR images depict the same content, this method enables the transfer of semantic information from the well-established RGB domain to SAR images. For *SARLANG-1M-VQA*, which primarily focuses on grounding and referring tasks for fine-grained SAR image understanding, we generate text descriptions directly from annotated bounding boxes in existing SAR datasets. This process constructs a novel text corpus, serving as a dedicated annotation resource for SAR VQA tasks.

a) SARLANG-1M-Cap: The SAR image captioning task involves generating a comprehensive textual description for a given SAR image, encapsulating intricate object details and contextual relationships. Unlike traditional classification tasks, this challenge requires predicting the global content of SAR images, ensuring that the generated captions capture not only the object categories present within the scene but also the spatial and semantic relationships among them.

Text Generation: SARLANG-1M-Cap is designed to provide detailed descriptions of SAR images across multiple dimensions, including global scene context, object-level information, and interobject relationships. To generate enriched and high-quality captions, we leverage three representative VLMs applied to paired RGB–SAR images.

1) *BLIP [76]:* It utilizes ViT-Large/16 [101] as its backbone and has been pretrained on 14 million images, including two human-annotated datasets (COCO [102] and Visual Genome [103]) and three web-scale datasets (Conceptual Captions [104], Conceptual 12M [104], and SBU Captions [105]).
2) *CLIP [67]:* Initially pretrained on the large-scale LAION-2B dataset [106], followed by fine-tuning on MSCOCO [102], making it well-suited for open-domain vision–language understanding.
3) *GPT-4o [16]:* It is the state-of-the-art multimodal LLM capable of generating context-aware and detailed image captions by leveraging its advanced vision–language reasoning capabilities. Unlike BLIP [76] and CLIP [67], GPT-4o [16] is a prompt-driven model for image captioning. In our implementation, we employ the prompt "Describe the content of this remote sensing image in one sentence." For caption generation. The specific model version used in this study is "gpt-4o-2024-08-06."

As shown in Fig. 5, BLIP [76] and CLIP [67] primarily generate concise captions, whereas GPT-4o [16] produces more complex and detailed descriptions. This diverse captioning approach enhances sentence richness and ensures that the generated text is not confined to a rigid or repetitive linguistic pattern.

b) SARLANG-1M-VQA: Given an SAR image, the VQA task involves answering questions based on the semantic understanding of the image content, as illustrated in Fig. 5. While *SARLANG-1M-Cap* emphasizes global-level understanding, *SARLANG-1M-VQA* is designed to place greater focus on local content comprehension—including the identification, localization, and reasoning about specific objects and regions within the SAR images.

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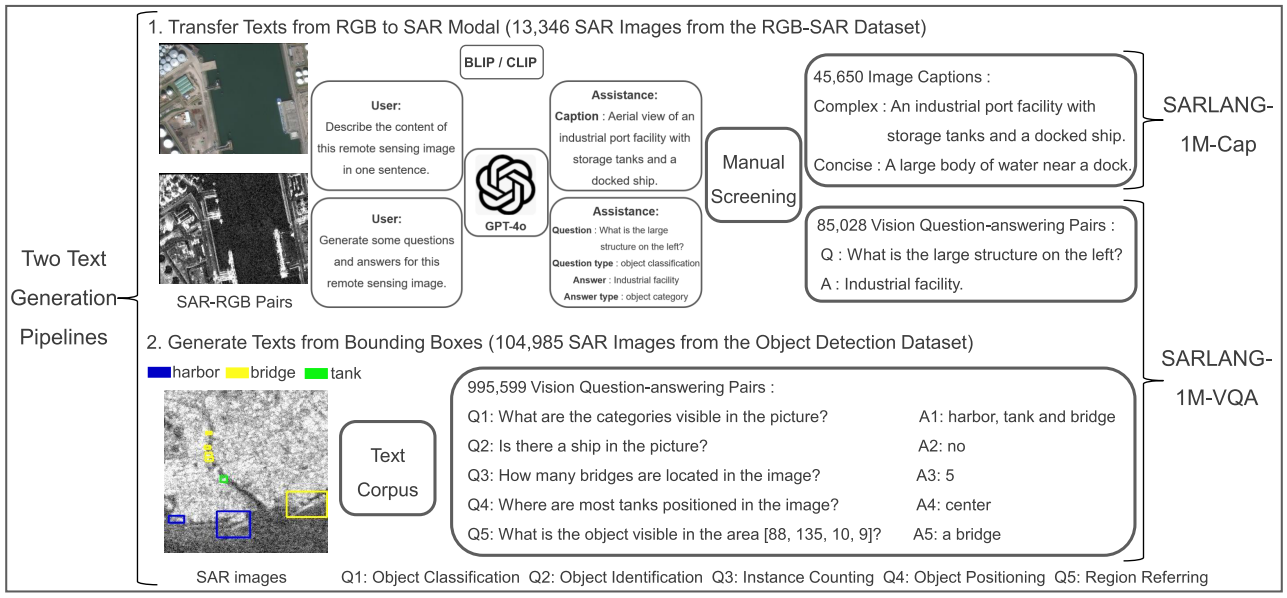


Fig. 5. Text annotation generation process of our *SARLANG-1M* dataset, including *SARLANG-1M-Cap* and *SARLANG-1M-VQA* benchmarks. Two text generation pipelines are adopted to construct our *SARLANG-1M* dataset. Notably, the manual screening is introduced to improve the quality of the generated text through the supervision of SAR experts. The text corpus is designed to generate text annotations from bounding boxes provided in the object detection dataset [18]. Objects in the blue box, yellow box, and green box refer to "harbor," "bridge," and "tank" categories, respectively. Q1–Q5 refer to the question types of object classification, object identification, instance counting, object positioning, and region referring, respectively.

TABLE V

TEXT CORPUS OF SARLANG-1M-VQA DATASET. THE TEMPLATES OF FIVE QUESTION TYPES AND CORRESPONDING ANSWERS ARE LISTED

Question Type	Question Template	Answer
Object Identification	Does the/Are there [specific category] exist in the image? Is there a [specific category] located on the [specific position] side of the image? Are there any other categories apart from [specific category]/in the image?	'Yes/No'
Object Classification	What is the main object/are the categories visible in the image?	'Ship', 'Aircraft', 'Car', 'Tank', 'Bridge', 'Harbor'
Instance Counting	How many [specific category] are visible/is located in the image?	Number
Object Positioning	Where is/are the most [specific category] positioned within the image?	'Left', 'Right', 'Top', 'Bottom', 'Center'
Region Referring	What is the object in the area [detailed coordinates]?	'Ship', 'Aircraft', 'Car', 'Tank', 'Bridge', 'Harbor'

Text Generation: To produce high-quality text annotations that provide precise quantifications and coordinate specifications for visible objects in SAR images, we introduce a novel text corpus as shown in Table V. Five primary question templates and corresponding answers are defined.

1) *Object Identification:* Designed to determine the presence of specific objects such as "Ship," "Tank," "Aircraft," "Bridge," "Car," and "Harbor," with answers being "yes" or "no."
2) *Object Classification:* It aims to identify the predominant category within the SAR image, with answers from the set ["Ship," "Tank," "Aircraft," "Bridge," "Car," "Harbor"].
3) *Instance Counting:* It investigates the quantity of a specific category within the SAR image, with the count as the answer.
4) *Object Positioning:* It determines the approximate location of a category, with answers being one of ["Left," "Right," "Top," "Bottom," "Center"].
5) *Region Referring:* It focuses on identifying the category within a specified area, with answers from the six potential categories ["Ship," "Tank," "Aircraft," "Bridge," "Car," "Harbor"].

To further diversify question–answer patterns and expand remote sensing categories beyond those in the SAR object detection dataset [18], we apply a similar modality transfer approach to generate VQA annotations. As shown in Table VI, various prompts are input to the GPT-4o [16] model, producing a wide range of captions and question–answer pairs and enhancing interpretative depth. In addition to enriching the text annotations for the five defined question types in our corpus, the remaining VQA labels constitute a "general inquiries" question type. This type of question explores novel remote sensing applications such as land cover classification, reasoning, and object shape prediction, as shown in Fig. 2. Leveraging two above text-generation pipelines, *SARLANG-1M-VQA* benchmark creates a comprehensive set of VQA annotations for four subdatasets [18], [83], [84], [85]. The detailed text statistics are presented in Table VII.

c) Quality Control: While the automated text generation pipeline provides a scalable approach to annotating SAR images, it is not without limitations. First, SAR images

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TABLE VI

INPUT PROMPTS FOR TEXT GENERATION WITH GPT-4O [16] MODEL

Text Type	Input Prompt
Caption	Describe the content of this remote sensing image in one sentence.
VQA	Generate some questions and answers for this remote sensing image, with the output format: question, answer, question type, answer type. Output some reasoning question-answer pairs for this remote sensing image, with the format: question, answer, question type, answer type. Output question-answer pairs based on the remote sensing objects in the image, with the format: question, answer, question type, answer type.

TABLE VII

TEXT STATISTICS FOR FOUR SUBDATASETS IN OUR SARLANG-1M DATASET

Benchmark	SpaceNet6 [83]	DFC2023 [84]	OpenEarthMap-SAR [85]	SARDet-100k [18]
SARLANG-1M-Cap	12,809	20,194	12,647	0
SARLANG-1M-VQA	23,176	31,528	30,324	995,599

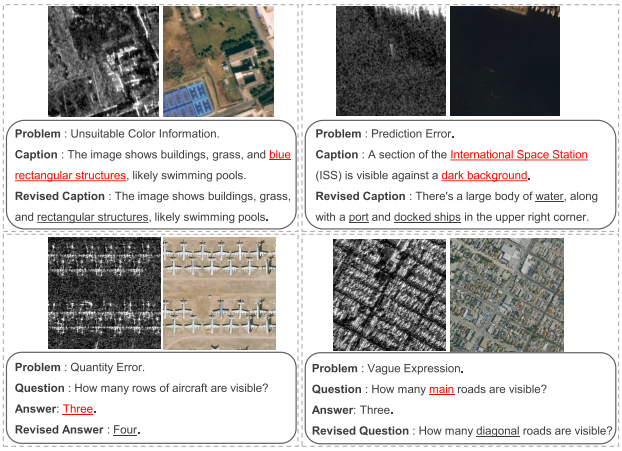


Fig. 6. Exemplar false cases before and after manual refining. Four types of cases are presented: unsuitable color information, prediction errors, quantity errors, and vague expressions.

TABLE VIII

STATISTICS FOR MANUAL SCREENING AND REVISION. "# REVISED TEXT" AND "# TOTAL TEXT" INDICATE THE NUMBER OF MANUALLY REVISED TEXTS AND THE TOTAL NUMBER OF TEXTS, RESPECTIVELY

Benchmark	# Revised Texts	# Total Texts	Proportion
SARLANG-1M-Cap	12,971	45,650	28.41%
SARLANG-1M-VQA	14,454	85,028	16.99%

inherently lack color information, leading to misalignment when descriptions include color-based attributes derived from paired RGB images. Second, the generated text annotations are not always accurate, as current VLMs have limited capability in fully comprehending remote sensing RGB imagery, which can result in errors when transferring descriptions to SAR images. To ensure high-quality textual annotations in *SARLANG-1M*, we implement a rigorous manual review and filtering process, conducted by domain experts. This ensures that incorrect, inconsistent, or irrelevant descriptions are identified and revised. The manual screening was conducted by three SAR experts, requiring approximately 310 person-hours in total. Two experts reviewed the generated descriptions to ensure their accuracy and relevance, while a third senior expert performed a cross-check and provided the final adjudication. Fig. 6 presents examples of failure cases and their corresponding corrections after expert verification. The revised text statistics during the manual screening process are shown in Table VIII.

IV. VLM EVALUATION FOR SAR

In this section, we present three VLM evaluation pipelines for SAR image understanding, including zero-shot testing, full fine-tuning, and parameter-efficient fine-tuning (PEFT). The VLM training and evaluation pipelines are demonstrated in Fig. 7.

A. Zero-Shot Testing

Zero-shot testing involves assessing VLMs on downstream tasks without any task-specific fine-tuning or additional training. This method evaluates the versatility and robustness of VLMs in executing SAR image understanding tasks, relying solely on the inherent knowledge acquired during initial pretraining. As illustrated in Fig. 7(a), the pretrained VLMs are directly utilized to process input SAR images, subsequently generating the output.

B. Full Fine-Tuning

Full fine-tuning entails adjusting all parameters of pretrained VLMs to adapt to a new, specific dataset or task. This process optimizes the model for the nuances of the target task, ensuring it refines its internal representations to effectively capture task-specific features. Given a VLM layer parameterized by $W_0 \in \mathbb{R}^{d\times k}$, to adapt it to a downstream task, we update it with $\Delta W \in \mathbb{R}^{d\times k}$, resulting in the updated layer parameterized by $W = W_0 + \Delta W$. For full fine-tuning, ΔW is computed based on all the parameters for the layer, which is computationally expensive and demands substantial GPU memory for VLMs.

C. Parameter-Efficient Fine-Tuning

PEFT technology enables the adaptation of VLMs to downstream tasks by fine-tuning only a small subset of (extra) model parameters. Based on the involvement of extra parameters, PEFT methods are categorized into extra-parameter [107], [108] and intraparameter methods [79], [109]. Extra-parameter methods preserve all original VLM parameters and introduce a set of learnable parameters to optimize the model's input or layers, as exemplified by adapter tuning [107] and prompt tuning [108]. In contrast, intraparameter methods

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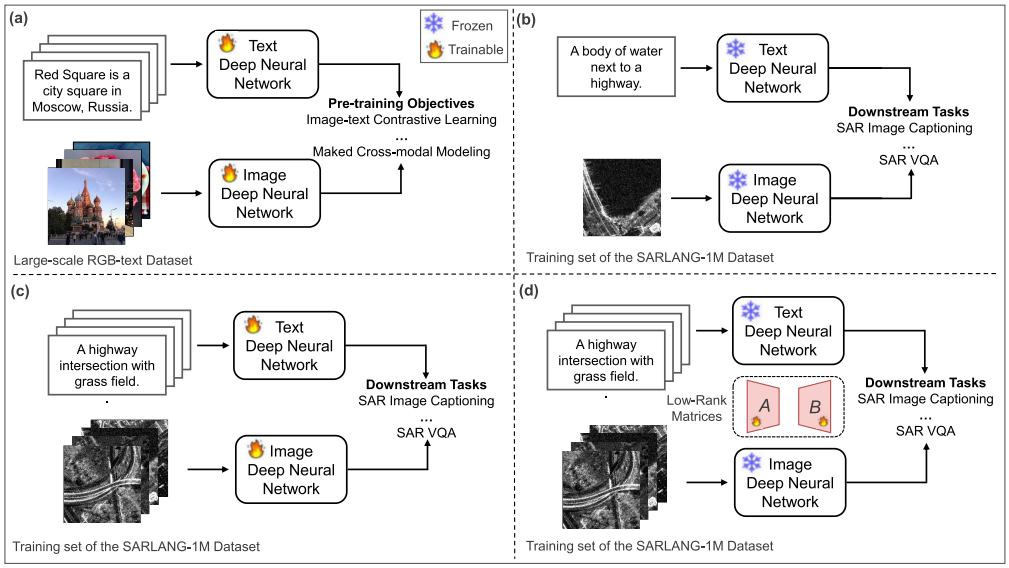


Fig. 7. VLM training and evaluation pipelines for SAR image understanding. (a) Pretraining. (b) Zero-shot. (c) Full fine-tuning. (d) LoRA [79] fine-tuning. As a typical PEFT technique, the LoRA [79] mechanism significantly reduces trainable parameters for downstream tasks by updating low-rank matrices while keeping the pretrained weight matrix fixed.

freeze most of the original VLM parameters, adjusting only a limited number. As a typical PEFT technology, the LoRA [79] mechanism focuses on keeping the pretrained weights fixed and constraining the update matrix ΔW to be low rank during fine-tuning, implemented via matrix decomposition

$$\Delta W = BA \tag{1}$$

where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and the rank $r \ll \min(d, k)$.

The LoRA fine-tuning process involves the following principles.

1) The original pretrained weights W_0 remain frozen and do not receive gradient updates during fine-tuning.
2) The low-rank matrices A and B are the only trainable parameters, capturing task-specific adjustments.
3) Both W_0 and ΔW are applied separately to the input vector x, and their outputs are combined.
4) The resulting output vectors are summed elementwise

$$f(x) = W_0 x + \Delta W x. \tag{2}$$

To ensure a fair and comprehensive evaluation of VLM's performance, we introduce two distinct evaluation methodologies based on the aforementioned VLM evaluation pipelines. First, we initialize the models with pretrained parameters from large-scale RGB image–text alignment datasets and directly evaluate their performance on the test set of the *SARLANG-1M* benchmark. This approach allows for an intuitive comparison of the VLMs' capabilities in SAR downstream tasks without additional fine-tuning (i.e., zero-shot testing). Second, we conduct the full fine-tuning and LoRA fine-tuning evaluation to fine-tune each VLM using the training set of the *SARLANG-1M* dataset and subsequently evaluate their performance on the test set of the *SARLANG-1M* benchmark. By comparing the results before and after fine-tuning, we quantitatively assess the contribution of the *SARLANG-1M* dataset to improving model performance on SAR downstream tasks.

TABLE IX
SAR–TEXT PAIR DISTRIBUTION STATISTICS IN OUR SARLANG-1M DATASET. "# TRAIN" AND "# TEST" DENOTE THE NUMBER OF TRAINING SAMPLES AND TESTING SAMPLES, RESPECTIVELY

Benchmark	Text	# Train	# Test
SARLANG-1M-Cap	Complex Caption	14,481	6,190
	Concise Caption	17,487	7,492
SARLANG-1M-VQA	VQA	955,372	125,255

V. EXPERIMENTAL ANALYSIS

In this section, some mainstream VLMs and non-VLMs are adopted to comprehensively evaluate our constructed *SARLANG-1M* dataset and demonstrate the effectiveness of our dataset. First, the experimental setup, including implementation details and evaluation metrics, is introduced. Then, extensive experiments on SAR image captioning and VQA tasks are conducted on our *SARLANG-1M* dataset with detailed analysis. We also provide some visual results to present the text quality of our dataset more intuitively. Finally, we provide ablation analyses on optional SAR image preprocessing strategies and justify their contributions in SAR image captioning and VQA tasks.

A. Experimental Setup

1) Implementation Details: In order to facilitate benchmark evaluation, the *SARLANG-1M* dataset has been divided into two distinct, nonoverlapping subsets, each specifically curated for model training and evaluation. To preserve the remote sensing category distribution of samples within the subset [18], the validation set is not established. The SAR–text pairs distribution statistics in our dataset are shown in Table IX. For the *SARLANG-1M-Cap* benchmark, following standard methodologies common in remote sensing image datasets [21], [110], all SAR–text samples are divided into training and test sets using a 7:3 ratio. This division resulted in a training set

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comprising 9341 images paired with 31 968 captions and a test set consisting of 4005 images accompanied by 13 682 captions, ensuring a robust foundation for VLM training and evaluation.

In the *SARLANG-1M-VQA* benchmark, for SAR samples identical to those in the *SARLANG-1M-Cap* benchmark, the same 7:3 ratio partitioning method for training and test sets has been employed. For the SAR images originating from the dataset [18], we preserved the original train/test image splits as established in the SARDet-100k dataset [18]. Consequently, the training component encompasses 103 834 SAR images paired with 955 372 corresponding textual annotations, while the evaluation subset includes 14 497 SAR images with 125 255 associated text annotations.

For the aforementioned benchmarks, we evaluate ten mainstream VLMs to demonstrate their potential in understanding SAR images and to highlight the contribution of our *SARLANG-1M* dataset. Among the evaluated VLMs, five models—BLIP [76], LLaVA1.5-7B [17], LLaVA1.5-13B [17], QWEN2-VL-7B [111], and QWEN2.5-VL-7B [19]—are fine-tuned using the LoRA [112] training method with a rank of 8 applied to all linear layers. This fine-tuning strategy ensures efficient adaptation while minimizing computational overhead. During the fine-tuning stage, each VLM is trained for 30 epochs with a batch size of 1. The learning rate is initialized at $1e^{-4}$ and incorporates a warm-up strategy with a ratio of 0.1. During the evaluation of these VLMs, the prompt for the SAR image captioning task is: "Describe the content of this image in one sentence," while the prompt for the SAR image VQA task is: "Question: {question}. Answer the question concisely."

For unified VLM processing, the single polarized SAR images in the DFC2023 [84], SARDet-100k [18], and OpenEarthMap-SAR [85] datasets are directly processed by VLMs. As for the quad-polarized SAR images in the SpaceNet6 [83] dataset, to ensure alignment with the OpenEarthMap-SAR [85] dataset, the amplitude information from the HH polarization bands is processed by VLMs. SAR image preprocessing is an optional strategy to improve the performance of VLMs by significantly enhancing image clarity and effectively highlighting key objects within the SAR images. The impact of SAR image preprocessing on VLMs' performance is discussed in Section V-C. The refined Lee filter [113] is utilized for denoising, while linear stretching [114] is applied after logarithmic transformation for contrast enhancement.

For the SAR image captioning task, two traditional models, namely, MLAT [54] and HCNet [55], are utilized as baseline networks to provide the comparison with VLMs. These traditional models are trained from scratch, adhering to their original specifications and hyperparameters. Additionally, they follow the same train/test split as the VLMs. All experiments are performed on two NVIDIA A100 GPUs. Our source code is publicly available to facilitate community replication and to support ongoing research in https://github.com/Jimmyxichen/SARLANG-1M

2) Evaluation Metrics: For SAR image captioning model evaluation, we adhere to standard practices established by existing VLMs [21], utilizing a comprehensive set of established metrics including traditional metrics (BLEU [115], ROUGE_L [116], and CIDEr [20]) and a GPT-4 based metric $CHAIR^2$ [117]. For the BLEU metric, we consider n-gram precision with n values of 1–4. A higher BLEU score indicates better performance and higher quality text outputs.

For the evaluation of SAR image VQA tasks, we employ a GPT-4 based metric overall accuracy [21] to assess the performance of existing VLMs on the *SARLANG-1M-VQA* benchmarks. In the GPT-4-based evaluation, we utilize GPT-4 to determine whether the predicted answers match the ground truth for each question. The prompt used is: "Question: {question}, Ground Truth Answer: {ground_truth}, Predicted Answer: {predicted answer}. Does the predicted answer match the ground truth? Answer 1 for match and 0 for not match. Use semantic meaning, not exact match. Synonyms are also treated as a match, e.g., oval and circular, pond and swimming pool." Overall accuracy is calculated as the ratio of the number of matches (1 s) to the total number of questions.

B. Benchmarking SARLANG-1M

1) SAR Image Captioning on SARLANG-1M-Cap Benchmark: As illustrated in Table X, the InternVL2.5-8B [118] model demonstrates significantly superior performance compared to other VLMs when evaluated on the test set without fine-tuning, achieving state-of-the-art results when evaluated against both complex and concise caption ground truths. This exceptional performance can be attributed to InternVL2.5-8B's [118] extensive pretraining on larger and more diverse datasets, which endows it with a comprehensive understanding of SAR image scenes and robust language generation capabilities. These advantages enable the InternVL2.5-8B [118] model to produce more descriptive and accurate captions.

After full fine-tuning on the *SARLANG-1M-Cap* benchmark, the LLaVA1.5-7B [17] model achieves the highest performance metrics when evaluated against complex caption ground truth, with a BLEU_1 score of 36.59, an ROUGE_L score of 33.53, and a CIDEr score of 44.88. On the contrary, LLaVA1.5-7B [17] underperforms relative to traditional models when evaluated against the concise caption ground truth, suggesting that the outputs of traditional models align more closely with simple, fixed-pattern descriptions. Specifically, the HCNet [55] model achieves the highest performance among traditional models, with a BLEU_1 score of 61.06, an ROUGE_L score of 60.78, and a CIDEr score of 239.00. The MLAT [54] model follows closely, securing the second-highest performance with a BLEU_1 score of 60.55, an ROUGE_L score of 60.03, and a CIDEr score of 224.00. Notably, the BLIP [76] model achieves the highest performance among all VLMs when evaluated against concise caption ground truth. This is primarily because most of the concise ground-truth captions in the *SARLANG-1M-Cap* benchmark are generated by the BLIP [76] model, resulting in a closer alignment between its outputs and the benchmark's concise caption patterns.

Furthermore, the performance improvements observed after LoRA fine-tuning on the *SARLANG-1M-Cap* benchmark validate the effectiveness of our dataset. Specifically, four VLMs—LLaVA1.5-7B [17], LLaVA1.5-13B [17], QWEN2-VL-7B [111], and QWEN2.5-VL-7B [19]—exhibit

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TABLE X

COMPARISON WITH EXISTING IMAGE CAPTIONING METHODS ON SARLANG-1M-CAP BENCHMARK. THE COMPLEX TEXT AND CONCISE TEXT REFER TO GROUND TRUTH GENERATED BY GPT-4O [16] MODEL AND BLIP [76]/CLIP [67] MODELS, RESPECTIVELY

Caption	Model	Param	Train Strategy	BLEU_1	BLEU_2	BLEU_3	BLEU_4	ROUGE_L	CIDEr
Complex	MLAT [54]	–	Train from Scratch	26.02	16.90	11.67	8.09	24.90	34.50
	HCNet [55]	–	Train from Scratch	26.31	16.57	11.12	7.42	24.91	33.61
	LLaVA1.5 [17]	13B	Zero-shot Testing	6.69	2.88	1.11	0.44	11.50	0.01
	LLaVA1.5 [17]	7B	Zero-shot Testing	7.13	3.08	1.11	0.44	12.06	0.03
	DeepSeek-VL [15]	7B	Zero-shot Testing	21.42	9.58	4.17	2.19	19.17	4.62
	DeepSeek-VL [15]	1.3B	Zero-shot Testing	22.35	8.23	3.69	1.89	18.88	5.80
	InternVL2.5 [118]	8B	Zero-shot Testing	28.11	17.90	12.19	7.51	29.66	16.15
	InternVL2.5 [118]	4B	Zero-shot Testing	27.96	18.28	12.60	7.89	29.98	15.38
	QWEN2-VL [111]	7B	Zero-shot Testing	6.48	2.78	0.96	0.35	11.33	0.01
	QWEN2.5-VL [19]	7B	Zero-shot Testing	24.56	13.69	8.87	5.35	24.49	9.42
	QWEN2.5-VL [19]	3B	Zero-shot Testing	26.70	15.31	9.04	4.14	26.06	8.96
	LLaVA1.5 [17]	7B	Full Fine-Tuning	**36.59**	**24.68**	**18.10**	**13.39**	**33.53**	44.88
	BLIP(Vit-Base) [76]	129M	LoRA Fine-Tuning	26.32	17.31	12.09	8.56	24.63	36.54
	LLaVA1.5 [17]	13B	LoRA Fine-Tuning	34.90	22.95	16.55	12.01	32.43	45.13
	LLaVA1.5 [17]	7B	LoRA Fine-Tuning	35.24	23.63	17.28	12.70	32.72	46.35
	QWEN2-VL [111]	7B	LoRA Fine-Tuning	35.78	23.72	17.57	13.08	32.84	48.36
	QWEN2.5-VL [19]	7B	LoRA Fine-Tuning	32.79	22.29	16.44	12.25	30.24	**55.64**
Concise	MLAT [54]	–	Train from Scratch	60.55	53.82	48.46	43.33	60.03	224.07
	HCNet [55]	–	Train from Scratch	**61.06**	**54.45**	**49.19**	**44.15**	**60.78**	**239.00**
	LLaVA1.5 [17]	13B	Zero-shot Testing	9.23	5.12	2.93	1.73	15.08	0.01
	LLaVA1.5 [17]	7B	Zero-shot Testing	9.22	4.84	2.26	1.07	14.72	0.02
	DeepSeek-VL [15]	7B	Zero-shot Testing	20.15	10.50	5.12	3.01	17.61	2.73
	DeepSeek-VL [15]	1.3B	Zero-shot Testing	24.98	14.88	6.33	3.45	22.60	3.78
	InternVL2.5 [118]	8B	Zero-shot Testing	29.50	20.72	14.93	9.85	30.28	4.84
	InternVL2.5 [118]	4B	Zero-shot Testing	26.58	17.50	12.02	7.47	26.89	4.46
	QWEN2-VL [111]	7B	Zero-shot Testing	8.38	4.27	2.07	1.02	14.07	0.01
	QWEN2.5-VL [19]	7B	Zero-shot Testing	18.42	7.85	3.90	1.57	17.68	2.85
	QWEN2.5-VL [19]	3B	Zero-shot Testing	32.17	23.10	17.33	12.17	30.97	5.48
	LLaVA1.5 [17]	7B	Full Fine-Tuning	22.04	11.77	7.38	4.76	19.51	9.74
	BLIP(Vit-Base) [76]	129M	LoRA Fine-Tuning	58.34	51.98	46.95	42.16	55.73	227.24
	LLaVA1.5 [17]	13B	LoRA Fine-Tuning	25.42	15.61	10.64	7.20	22.65	11.03
	LLaVA1.5 [17]	7B	LoRA Fine-Tuning	23.97	14.09	9.00	5.71	21.52	10.27
	QWEN2-VL [111]	7B	LoRA Fine-Tuning	24.58	15.14	10.41	7.11	22.28	9.56
	QWEN2.5-VL [19]	7B	LoRA Fine-Tuning	37.75	31.00	26.31	22.36	38.50	70.05

significant enhancements in performance metrics. Among these, QWEN2-VL-7B [111] demonstrates the most substantial improvements for the complex captions, with enhanced scores of 29.30 for BLEU_1, 21.51 for ROUGE_L, and 48.35 for CIDEr. As for the concise captions, QWEN2.5-VL-7B [19] achieves the highest performance improvements, with enhanced scores of 19.33 for BLEU_1, 20.82 for ROUGE_L, and 67.20 for CIDEr. These results underscore the value of the *SARLANG-1M-Cap* benchmark in advancing the VLM's capabilities in the SAR image captioning task.

While traditional metrics like BLEU [115] rely on n-gram overlap with few references and thus miss compositional semantics in complex captions, we also utilize a GPT-4-based metric named CHAIR^2 [117] for a more robust evaluation of the complex texts. Table XI reports the SAR image captioning performance in terms of CHAIR^2 [117]. As shown in Table XI, the significant performance enhancements were observed for the LLaVA1.5-7B [17], QWEN2-VL-7B [111], and QWEN2.5-VL-7B [19] models after LoRA fine-tuning, with CHAIR^2 [117] enhancements of 33.83, 44.16, and 50.97, respectively. These results further underscore the value of the *SARLANG-1M-Cap* benchmark for enhancing VLMs' ability to interpret SAR images.

To offer an intuitive comparison, we present a visualization of selected image captioning results from the *SARLANG-1M-Cap* benchmark in Fig. 8. Notably, when evaluating the pretrained LLaVA1.5-7B [17] model on the *SARLANG-1M-Cap* benchmark, the model misclassifies the scene as "a night scene" and incorrectly identifies the key objects "storage tanks" as "bright lights." This misclassification arises because the pretrained LLaVA1.5-7B [17] model lacks sufficient knowledge of the SAR image. However, after LoRA fine-tuning on the *SARLANG-1M-Cap* benchmark, the LLaVA1.5-7B [17] model demonstrates significant improvement, accurately identifying the scene and generating captions that include the correct keyword "infrastructure." Additionally, the QWEN2-VL-7B [111] model demonstrates comparable improvements, accurately identifying key elements such as "industrial facility" and "containers" within the SAR image. These results highlight the effectiveness of the *SARLANG-1M-Cap* benchmark in enhancing the VLM's capability to interpret SAR image. Besides, we compare the text captions

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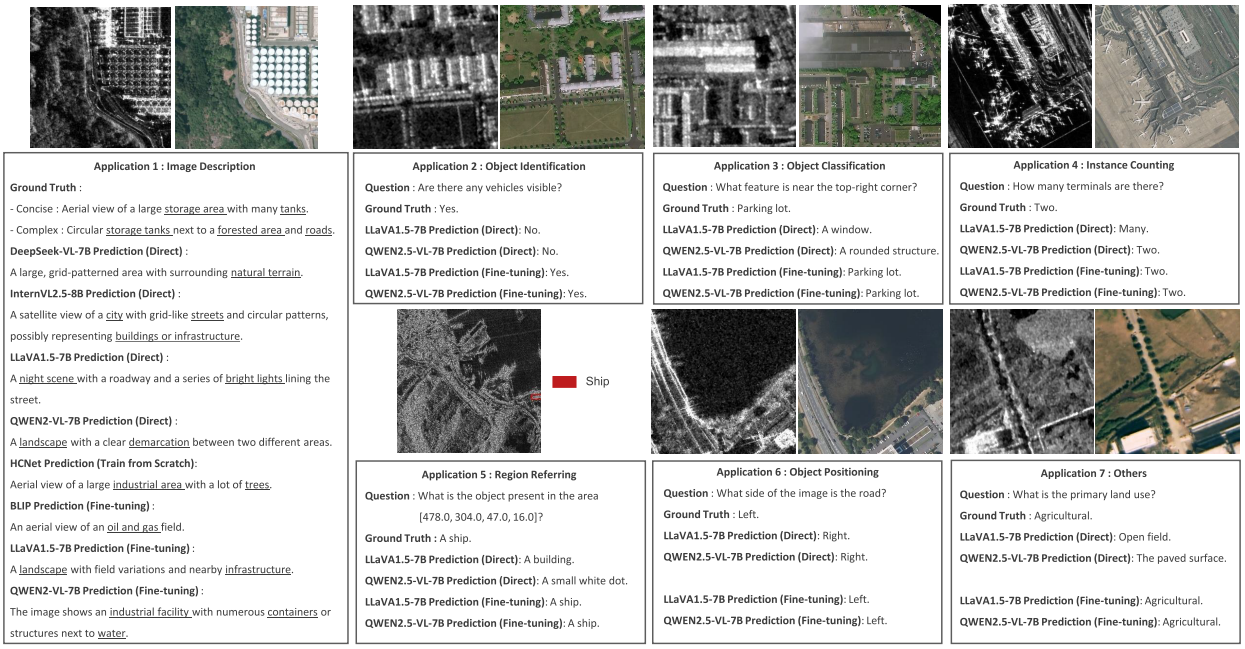


Fig. 8. Visualization results of multiple VLMs' predictions on our *SARLANG-1M* benchmark. The visualization results are presented across seven distinct application scenarios. "Direct" denotes the zero-shot testing, while "Fine-tuning" represents the evaluation after model LoRA fine-tuning on the *SARLANG-1M* dataset. "Train from Scratch" means the evaluation after model training from scratch.

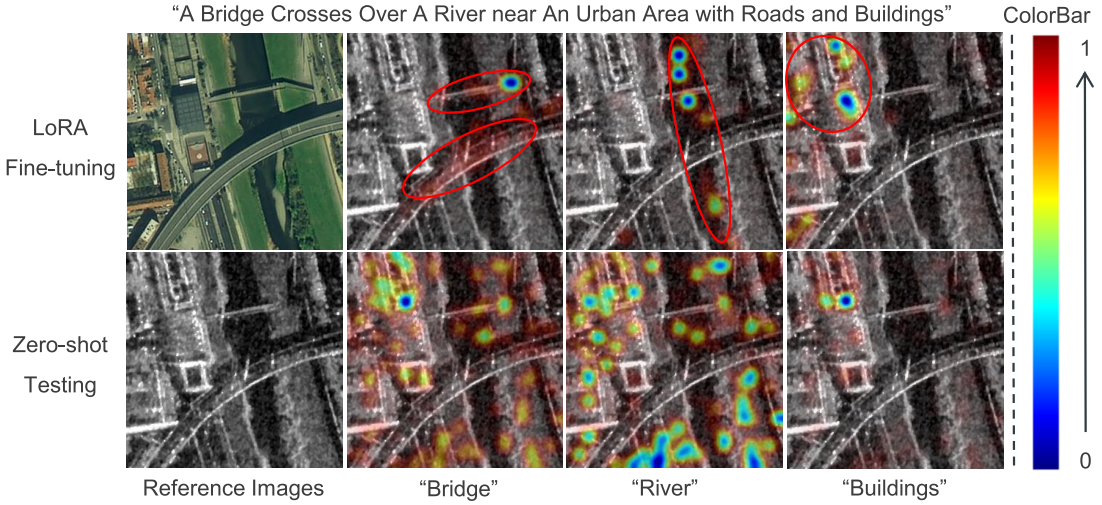


Fig. 9. Per-word Grad-CAM visualizations predicted by the BLIP model before and after fine-tuning on our *SARLANG-1M-Cap* benchmark. The text predicted by the BLIP [76] model, along with the key targets identified within the text, is marked above and below the image, respectively. "Zero-shot Testing" and "LoRA Fine-tuning" denote the prediction results generated by the BLIP model before and after LoRA fine-tuning on our *SARLANG-1M-Cap* dataset, respectively. Regions with strong responses to the given words are highlighted in red, while those with weak responses are highlighted in blue. After fine-tuning on our *SARLANG-1M-Cap* dataset, the BLIP [76] model improves in identifying areas corresponding to the given words, particularly those enclosed in red circles.

generated by HCNet [55], BLIP [76], DeepSeek-VL-7B [15], and InternVL2.5-8B [118]. The HCNet [55] model demonstrates the ability to generate captions that accurately incorporate key terms such as "industrial area" and "trees." Similarly, the BLIP [76] model produces captions containing the correct phrase "oil and gas field." In contrast, the DeepSeek-VL-7B [15] and InternVL2.5-8B [118] models exhibit misclassifications, describing storage tanks as a "grid-patterned area" and "grid-like streets," respectively. These results reveal that models fine-tuned on the *SARLANG-1M* dataset consistently generate more accurate and detailed captions. In comparison, models without fine-tuning on *SARLANG-1M* tend to produce captions that include incorrect or ambiguous terminology.

Furthermore, to investigate the correlation between the generated texts and SAR images, we constructed a per-word Grad-CAM [22] visualization to illustrate the response levels between key vocabulary in the captions generated by the BLIP [76] model and key objects in the SAR images. In comparison with the results depicted in the second row in Fig. 9, the results presented in the first row demonstrate that the *SARLANG-1M-Cap* dataset significantly enhances the BLIP [76] model's capability to accurately identify and localize key objects within SAR images.

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TABLE XI

COMPARISON WITH EXISTING IMAGE CAPTIONING METHODS ON THE SARLANG-1M-CAP BENCHMARK FOR THE COMPLEX TEXT EVALUATION

Caption	Model	Param	Train Strategy	CHAIR^2
Complex	MLAT [54]	–	Train from Scratch	66.46
	HCNet [55]	–	Train from Scratch	**67.19**
	LLaVA1.5 [17]	13B	Zero-shot Testing	18.16
	LLaVA1.5 [17]	7B	Zero-shot Testing	19.83
	DeepSeek-VL [15]	7B	Zero-shot Testing	25.36
	DeepSeek-VL [15]	1.3B	Zero-shot Testing	16.52
	InternVL2.5 [118]	8B	Zero-shot Testing	**40.36**
	InternVL2.5 [118]	4B	Zero-shot Testing	37.09
	QWEN2-VL [111]	7B	Zero-shot Testing	13.45
	QWEN2.5-VL [19]	7B	Zero-shot Testing	14.59
	QWEN2.5-VL [19]	3B	Zero-shot Testing	32.73
	LLaVA1.5 [17]	7B	Full Fine-Tuning	56.09
	BLIP(Vit-Base) [76]	129M	LoRA Fine-Tuning	63.18
	LLaVA1.5 [17]	13B	LoRA Fine-Tuning	51.40
	LLaVA1.5 [17]	7B	LoRA Fine-Tuning	53.66
	QWEN2-VL [111]	7B	LoRA Fine-Tuning	57.61
	QWEN2.5-VL [19]	7B	LoRA Fine-Tuning	**65.56**

TABLE XII

COMPARISON WITH EXISTING IMAGE VQA METHODS ON SARLANG-1M-VQA BENCHMARK. THE ACCURACY METRIC DENOTES THE RATIO OF CORRECTLY ANSWERED QUESTIONS

Model	Param	Train Strategy	Accuracy
LLaVA1.5 [17]	13B	Zero-shot Testing	29.91
LLaVA1.5 [17]	7B	Zero-shot Testing	33.55
QWEN2-VL [111]	7B	Zero-shot Testing	28.33
DeepSeek-VL [15]	7B	Zero-shot Testing	29.26
DeepSeek-VL [15]	1.3B	Zero-shot Testing	30.02
InternVL2.5 [118]	8B	Zero-shot Testing	39.90
InternVL2.5 [118]	4B	Zero-shot Testing	**43.95**
QWEN2.5-VL [19]	7B	Zero-shot Testing	38.60
QWEN2.5-VL [19]	3B	Zero-shot Testing	41.23
LLaVA1.5 [17]	7B	Full Fine-Tuning	71.08
LLaVA1.5 [17]	13B	LoRA Fine-Tuning	70.04
LLaVA1.5 [17]	7B	LoRA Fine-Tuning	70.30
QWEN2-VL [111]	7B	LoRA Fine-Tuning	68.55
QWEN2.5-VL [19]	7B	LoRA Fine-Tuning	**73.33**

2) SAR VQA on SARLANG-1M-VQA Benchmark: Table XII presents the VQA performance of various VLMs on our *SARLANG-1M-VQA* dataset. As demonstrated in Table XII, among the models evaluated without fine-tuning on our *SARLANG-1M-VQA* training set, the InternVL2.5-4B [118] model achieves state-of-the-art performance, attaining an overall accuracy of 43.95%. Besides, the QWEN2.5-VL-7B [19] model, after LoRA fine-tuning on the *SARLANG-1M-VQA* training set, achieves the highest overall accuracy of 73.33%. Furthermore, the baseline LLaVA1.5-7B [17] model with LoRA fine-tuning and full fine-tuning on our *SARLANG-1M-VQA* training set gets an overall accuracy of 70.30% and 71.08% on our *SARLANG-1M-VQA* test set, respectively. Compared to the performance before fine-tuning, the LLaVA1.5-7B [17] model exhibited a significant improvement in overall accuracy, with an increase of 37.53% and 36.75% after full fine-tuning and LoRA fine-tuning, respectively. Similar performance improvements were observed for the QWEN2-VL-7B [111], QWEN2.5-VL-7B [19], and LLaVA1.5-13B [17] models after LoRA fine-tuning, with overall accuracy enhancements of 40.22%, 34.73%, and 40.13%, respectively. These results demonstrate that our dataset substantially enhances the performance of VLMs on the SAR VQA task.

TABLE XIII

COMPARISON BETWEEN HUMAN EVALUATION AND VLM PERFORMANCE ON THE SARLANG-1M-VQA VALIDATION SET (300 SAMPLES). THE ACCURACY METRIC DENOTES THE RATIO OF CORRECTLY ANSWERED QUESTIONS. "–" REFERS TO NOT APPLICABLE

Model	Param	Train Strategy	Accuracy
QWEN2-VL [111]	7B	Zero-shot Testing	27.33
DeepSeek-VL [15]	7B	Zero-shot Testing	28.33
DeepSeek-VL [15]	1.3B	Zero-shot Testing	30.66
InternVL2.5 [118]	8B	Zero-shot Testing	39.00
InternVL2.5 [118]	4B	Zero-shot Testing	**41.66**
QWEN2.5-VL [19]	7B	Zero-shot Testing	35.66
QWEN2.5-VL [19]	3B	Zero-shot Testing	38.66
QWEN2-VL [111]	7B	LoRA Fine-Tuning	63.66
QWEN2.5-VL [19]	7B	LoRA Fine-Tuning	**72.66**
Ordinary People	–	–	20.55
SAR Experts	–	–	**62.66**

To further evaluate the SAR image interpretation capability of VLMs, we compared their performance against human participants. Given the large-scale nature of *SARLANG-1M*, we randomly selected 300 question–answer pairs from the *SARLANG-1M-VQA* test set to form a *validation* subset for human evaluation. Three SAR experts and three nonexpert participants were invited to answer these questions.

The performance comparison between various VLMs and the averaged human accuracies is presented in Table XIII. As shown, after fine-tuning with *SARLANG-1M*, the QWEN2.5-VL-7B [19] and QWEN2-VL-7B [111] models exhibit significant improvements in SAR interpretation and even outperform SAR experts on this *validation* set. These results demonstrate the great potential of VLMs to overcome nonexpert barriers in SAR image interpretation and to achieve a level of understanding comparable to human experts.

We also present a visualization of selected SAR VQA results from the *SARLANG-1M-VQA* benchmark in Fig. 8. The experimental results demonstrate significant performance improvements for the LLaVA1.5-7B [17] model and QWEN2.5-VL-7B [19] model across applications 2–7 after LoRA fine-tuning. A notable example is observed in application 4, where the pretrained LLaVA1.5-7B [17] model fails to accurately quantify targets in SAR images. However, after LoRA fine-tuning on our dataset's training set, the LLaVA1.5-7B [17] model achieves precise quantification of the "terminal" target class. Besides, application 7 addresses a fundamental remote sensing task: land cover classification. When processing SAR images of agricultural areas dominated by crops with sparse distributions of trees and grasslands, the pretrained LLaVA1.5-7B [17] model and QWEN2.5-VL-7B [19] model initially classify the entire region as "open field" and "a small white dot," respectively. Following LoRA fine-tuning on our

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TABLE XIV

SAR IMAGE CAPTIONING RESULTS ON SARLANG-1M-CAP BENCHMARK. "PRE" REPRESENTS THE PREPROCESSING STRATEGIES, WHILE "✓/✗" DENOTES THE MODEL EVALUATION WITH/WITHOUT SAR IMAGE PREPROCESSING

Caption	Model	Param	Pre	BLEU_1	BLEU_2	BLEU_3	BLEU_4	ROUGE_L	CIDEr
Complex	QWEN2.5-VL [19]	3B	✗	26.09	14.92	9.87	6.11	25.31	12.40
	QWEN2.5-VL [19]	3B	✓	**26.95**	**16.44**	**11.10**	**6.65**	**26.45**	**14.78**
Concise	QWEN2.5-VL [19]	3B	✗	25.18	15.57	10.84	7.35	23.73	5.79
	QWEN2.5-VL [19]	3B	✓	**30.57**	**21.54**	**15.97**	**10.97**	**29.36**	**6.25**

TABLE XV

SAR VQA RESULTS ON SARLANG-1M-VQA BENCHMARK. "PRE" REPRESENTS THE PREPROCESSING STRATEGIES, WHILE "✓/✗" DENOTES THE MODEL EVALUATION WITH/WITHOUT SAR IMAGE PREPROCESSING. THE ACCURACY METRIC DENOTES THE RATIO OF CORRECTLY ANSWERED QUESTIONS

Model	Param	Pre	Accuracy
QWEN2.5-VL [19]	3B	✗	29.51
QWEN2.5-VL [19]	3B	✓	**32.32**

SARLANG-1M-VQA dataset, the LLaVA1.5-7B [17] model and the QWEN2.5-VL-7B [19] model demonstrate enhanced capability, accurately identifying the primary land use patterns. These experimental results demonstrate that our *SARLANG-1M-VQA* dataset significantly enhances the performance of existing VLMs on SAR image understanding tasks.

C. Preprocessing

In this section, we analyze the impacts of SAR preprocessing operations, specifically denoising and contrast stretching, on the performance of VLMs in SAR image captioning and VQA tasks. The QWEN2.5-VL-3B [19] model, without fine-tuning on our *SARLANG-1M* dataset, is chosen for evaluation.

1) SAR Image Captioning on SARLANG-1M-Cap Benchmark: To demonstrate the effectiveness of preprocessing operations on the SAR image captioning task, we conduct two comparative experiments to quantify the contribution of the SAR image preprocessing strategy. We selected 1170 complex captions and 2571 concise captions from the test set of the *SARLANG-1M-Cap* benchmark, serving as ground truth for SAR images from the OpenEarthMap-SAR [85] dataset. As shown in Table XIV, preprocessing the SAR images significantly enhances the performance of the QWEN2.5-VL-3B [19] model, particularly for concise captions. Specifically, the model achieves improved scores of 5.39 for BLEU_1, 5.63 for ROUGE_L, and 0.46 for CIDEr. These results confirm the effectiveness of SAR image preprocessing operations on the VLM's performance in the SAR image captioning task.

2) SAR VQA on SARLANG-1M-VQA Benchmark: To further confirm the effectiveness of preprocessing operations in the SAR VQA task, 9220 VQA labels are selected from the test set of the *SARLANG-1M-VQA* benchmark, serving as ground truth for SAR images from the OpenEarthMap-SAR [85] dataset. As shown in Table XV, our SAR image preprocessing operations result in an overall accuracy improvement of 2.81 for the QWEN2.5-VL-3B [19] model. The performance improvements are achieved by the denoising operation, which removes elements like speckle noise that hinder VLMs' comprehension of SAR images. Additionally, linear stretching enhances SAR image contrast, enabling VLMs to focus more effectively on salient objects and areas.

VI. LIMITATIONS

While the *SARLANG-1M* dataset provides valuable, large-scale, and high-quality data for advancing SAR vision–language modeling, it still has several potential limitations that may affect its broader applicability toward general artificial intelligence for SAR interpretation. The first limitation concerns data coverage. The dataset is primarily collected from 59 cities worldwide, and therefore, the diversity of urban types and remote sensing targets remains limited. Expanding the geographic and thematic diversity of the dataset could further enhance its representativeness and generalization capability. The second limitation involves the polarization diversity of the SAR imagery. Among the source datasets, only SpaceNet6 [83] provides quad-polarization data (HH, VV, HV, and VH). The scarcity of multipolarization SAR data currently constrains the benchmark's capacity to evaluate model performance across different polarization modes. In future work, we plan to incorporate more geographically and physically diverse SAR data to further enhance the comprehensiveness and generalizability for broader SAR understanding tasks.

VII. CONCLUSION

In this article, we introduced *SARLANG-1M*, a large-scale dataset designed to advance multimodal understanding of SAR images. *SARLANG-1M* comprises approximately 1 million high-quality SAR image–text pairs collected from over 59 cities worldwide, offering multiscale resolutions ranging from 0.1 to 25 m, fine-grained semantic descriptions (including both concise and complex captions), and diverse remote sensing categories (1696 object categories and 16 land cover categories). The dataset features two benchmarks, *SARLANG-1M-Cap* and *SARLANG-1M-VQA*, which support seven key remote sensing applications, including image description, object identification, object classification, instance counting, region referring, object positioning, and others. Extensive experiments demonstrate that fine-tuning existing VLMs on *SARLANG-1M* significantly enhances their performance in SAR image understanding tasks, achieving interpretative abilities comparable to human experts. Further, our SAR image preprocessing strategies effectively enhance the VLMs' ability in SAR image interpretation.

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APPENDIX A
OVERALL IMAGE SIMILARITY

Given a image set $[f_1, f_2, \ldots, f_N]$ with N images, the *overall image similarity* $S_{\text{Image}} = (\mu, \sigma)$ is computed as follows:

$$V_i = \text{Feature_Extractor}\left[f_i\right] \tag{3}$$

$$V_t = \text{Feature_Extractor}\left[f_t\right] \tag{4}$$

where V_i and V_t correspond to the extracted image features of f_i and f_t. The VGG16 [94] model is designated as Feature_Extractor.

Following image feature extraction, the cosine similarity between each image pair (f_i, f_t) is then calculated based on the extracted features(V_i, V_t)

$$S_{i,t} = \frac{V_i \cdot V_t}{||V_i||||V_t||} \tag{5}$$

where $\cdot$ denotes the dot product, and $\|\cdot\|$ represents the Euclidean norm of the feature vectors.

The cosine similarities $S_{i,t}$ computed for all image pairs form an image similarity matrix S. To prevent redundant calculations, only the lower triangle of S is retained, creating an image similarity set s

$$s = \{S_{\text{it}} \mid i > t\}. \tag{6}$$

Then, we compute the mean μ and variance σ^2 of the similarity set s to represent the *overall image similarity* S_{Image} of the dataset

$$\mu = \frac{1}{M}\sum_{k=1}^{M} s_k \tag{7}$$

$$\sigma^2 = \frac{1}{M}\sum_{k=1}^{M} (s_k - \mu)^2 \tag{8}$$

$$S_{\text{Image}} = (\mu, \sigma) \tag{9}$$

where M indicates the number of image pairs.

APPENDIX B
OVERALL TEXT SIMILARITY

Given a text set $[h_1, h_2, \ldots, h_P]$ with P text, the *overall text similarity* $S_{\text{Text}} = (\mu_1, \sigma_1)$ is defined as follows:

$$E_j = \text{Feature_Extractor}\left[h_j\right] \tag{10}$$

$$E_k = \text{Feature_Extractor}\left[h_k\right] \tag{11}$$

where E_j and E_k are the extracted features of the texts h_j and h_k, respectively. The Sentence-BERT [95] model is denoted as Feature_Extractor. The cosine similarity between the text embeddings text pairs (h_j, h_k) is calculated as

$$L_{j,k} = \frac{E_j \cdot E_k}{||E_j||||E_k||}. \tag{12}$$

Then, the lower triangle of the text similarity matrix L constitutes a text similarity set l

$$l = \left\{L_{jk} \mid j > k\right\}. \tag{13}$$

The mean μ_1 and variance σ_1^2 are calculated from the similarity set l of text pairs

$$\mu_1 = \frac{1}{M_1}\sum_{k=1}^{M_1} s1_k \tag{14}$$

$$\sigma_1^2 = \frac{1}{M_1}\sum_{k=1}^{M_1} (l_k - \mu_1)^2 \tag{15}$$

$$S_{\text{Text}} = (\mu_1, \sigma_1) \tag{16}$$

where M_1 represents the number of text pairs.

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